

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

Sub: PRINCIPLES OF ELECTRICAL ENGINEERING

Theory QUESTIONS (Mo 15)

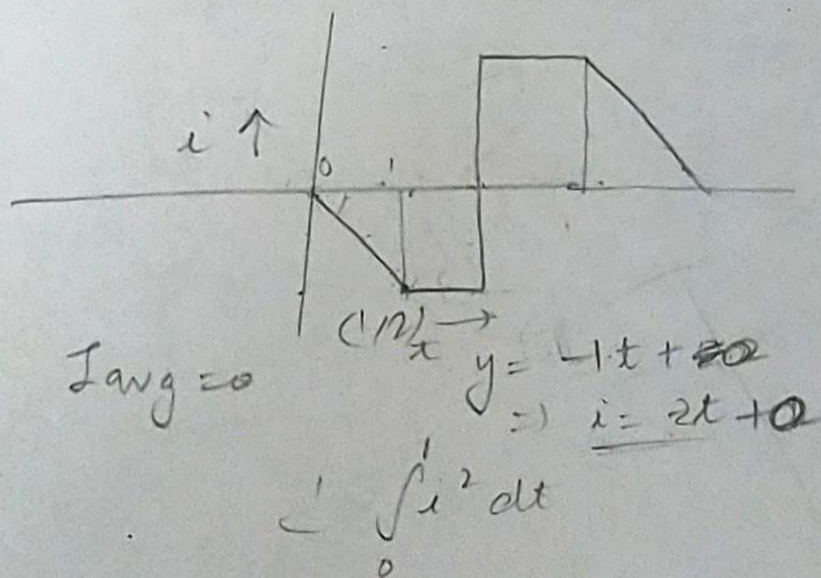
1. Define RMS value of an alternating current. Explain why this value is more generally employed in ac measurements than either the average value or peak value.
2. Explain the method of circulating rms value for non-sinusoidal and sinusoidal alternating quantity.
3. Explain with the aid of a sketch, how the average and rms values of an alternating current are obtained.
4. Derive an expression for instantaneous value of alternating current in terms of its maximum value, angular velocity and time.
5. Define the term inductive/capacitive reactance and explain its variation with frequency.
6. Prove that the average power in an ac circuit is equal to $VI\cos\phi$. Explain the significance of $\cos\phi$ in the expression.
7. Explain briefly the phenomenon of electrical resonance in ac circuits.
8. What happens to the power factor of a circuit when the frequency of the supply is varied above and below the resonant frequency?
9. Discuss the effects of varying the frequency upon the current drawn and the power factor in a series R-L-C circuit.
10. Explain with the aid of phasor diagram, the phenomenon of resonance in a circuit containing an inductor, a capacitor and a resistor in series.
11. Deduce an expression for resonant frequency of a series ac circuit.
12. What is meant by Q-factor of a tuned circuit?
13. Show that the resonant frequency of a series R-L-C circuit is the geometric mean of lower and upper half-power frequencies.
14. For a series R-L-C circuit, determine the half-power frequencies ω_1 and ω_2 in terms of quality factor Q_0 and the resonant frequency ω_0 .
15. Deduce the relationship between Q and bandwidth about the frequency of resonance.
16. Define and explain voltage magnification in the context of series resonance.
17. Write down the properties of a series 2nd order series/parallel R-L-C resonant circuit.
18. (a) With the aid of sketch, show variation of resistance, inductive and capacitive reactance, impedance, power factor, current versus frequency (in case of series circuit).
(b) With the aid of sketch, show variation of conductance, susceptance, admittance, power factor, current versus frequency (in case of parallel circuit).
19. Write notes on: (i) Supermesh analysis, (ii) Supernode analysis of solving a circuit.
20. Write notes on: dependent and independent sources.
21. State and explain KCL and KVL. Also mention the limitations of Kirchhoff's laws.
22. Give reasons for your choice of mesh current method and node voltage method of circuit analysis.
23. Differentiate between emf and terminal voltage in a circuit.
24. With the help of graph, explain the behaviour of an ideal and practical voltage source and current source.
25. Derive expressions for star to delta and delta to star transformations.
26. What is the value of the internal resistance of an ideal voltage source? Explain. ∞
27. What is the value of the internal resistance of an ideal current source? Explain. 0
28. When an ideal current source is placed as part of a closed circuit, comment on the factors that determines the voltage that appears across the current source terminals.

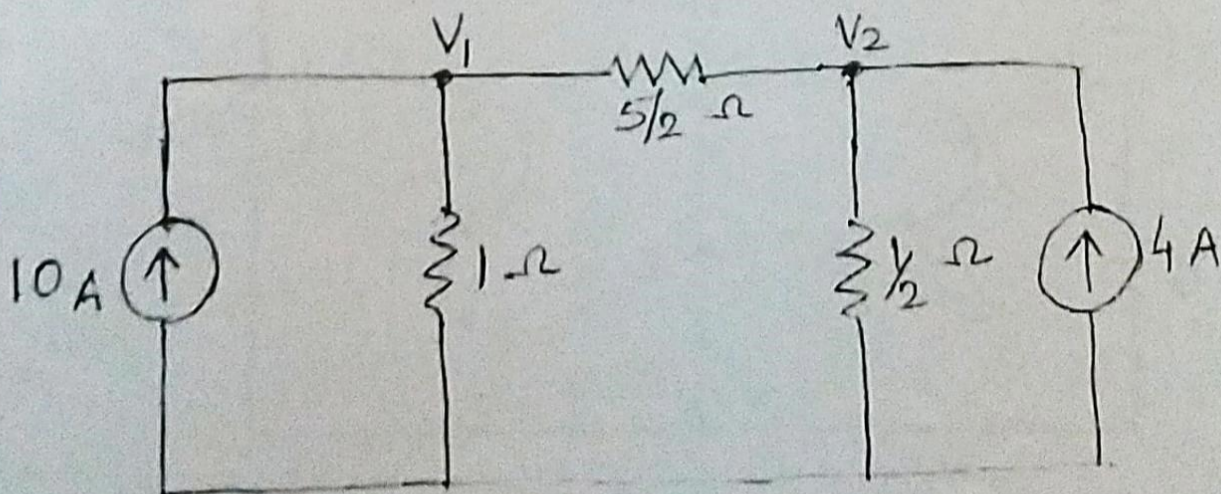
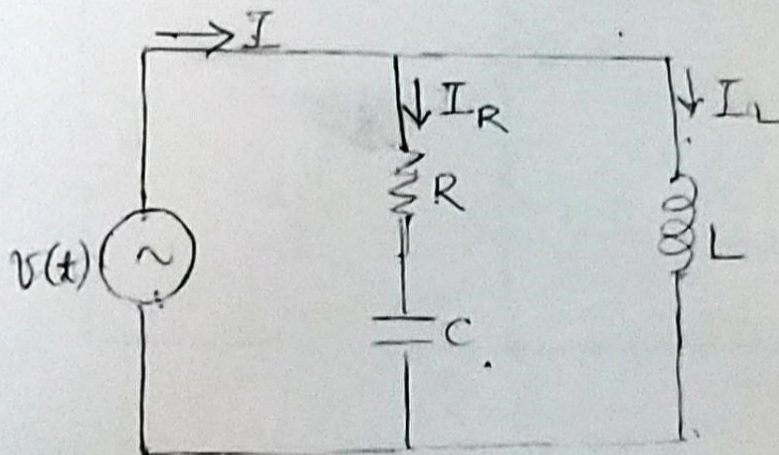
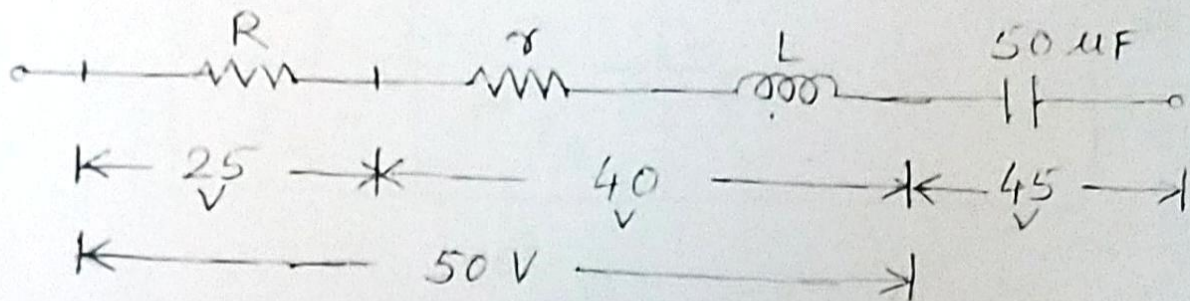
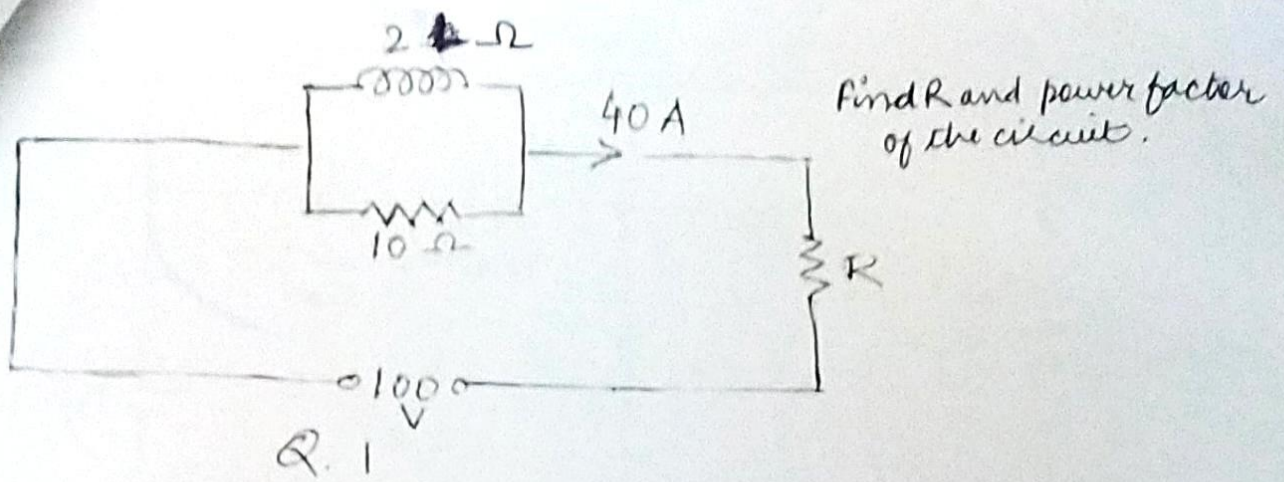
29. When an ideal current source is placed as part of a closed circuit, comment on the factors that determines the magnitude of the current associated with the voltage source.
30. Describe the convention used to define the positive direction for a voltage drop that appears across a passive circuit element in a closed circuit.
31. What is a ground reference? What is its purpose in an electric circuit?
32. Describe and illustrate the voltage divider principle.
33. Describe and illustrate the current divider principle.
34. Distinguish among a mesh current, a branch current and a loop current.
35. Why are mesh currents important in circuit analysis?
36. Distinguish between a node and an independent node.
37. Distinguish between a mesh and a loop of a circuit.
38. In a network, it is possible to identify four meshes and four independent nodes, which method of circuit analysis is to be preferred? Why?
39. A network is found to have a total of seven branches and four junctions points, which method is to be preferred in analysis of the circuit? Explain.
40. Define power factor and explain why it arrives in sinusoidal A.C. circuits?
41. Distinguish between instantaneous power and average power in a sinusoidal A.C. circuit?
42. What is apparent power? What practical significance does it have?
43. Distinguish among admittance, susceptance, conductance. What is complex admittance?
44. Draw the phasor diagram of a series RLC circuit using source voltage as the reference phasor and assuming that the corresponding line diagram of the source voltage and response current.
45. Can the phenomenon of resonance occur in engineering situation when a signal energy-storing element is present? Explain.
46. In a series RLC circuit, identify the character of the complex impedance when the circuit is driven at a frequency that exceeds the resonant frequency?
47. How is the current response related to voltage source at resonant frequency?
48. How is the Q-factor of a series RLC circuit defined? Of what practical significance is this number? Explain.
49. In a series RLC circuit, how is the voltage across inductor coil related to the applied source voltage in magnitude and phase, at resonant frequency?
50. How is the bandwidth of a series RLC circuit defined? Upon which parameters does the bandwidth depend? What role do the remaining circuit parameters play in this consideration?

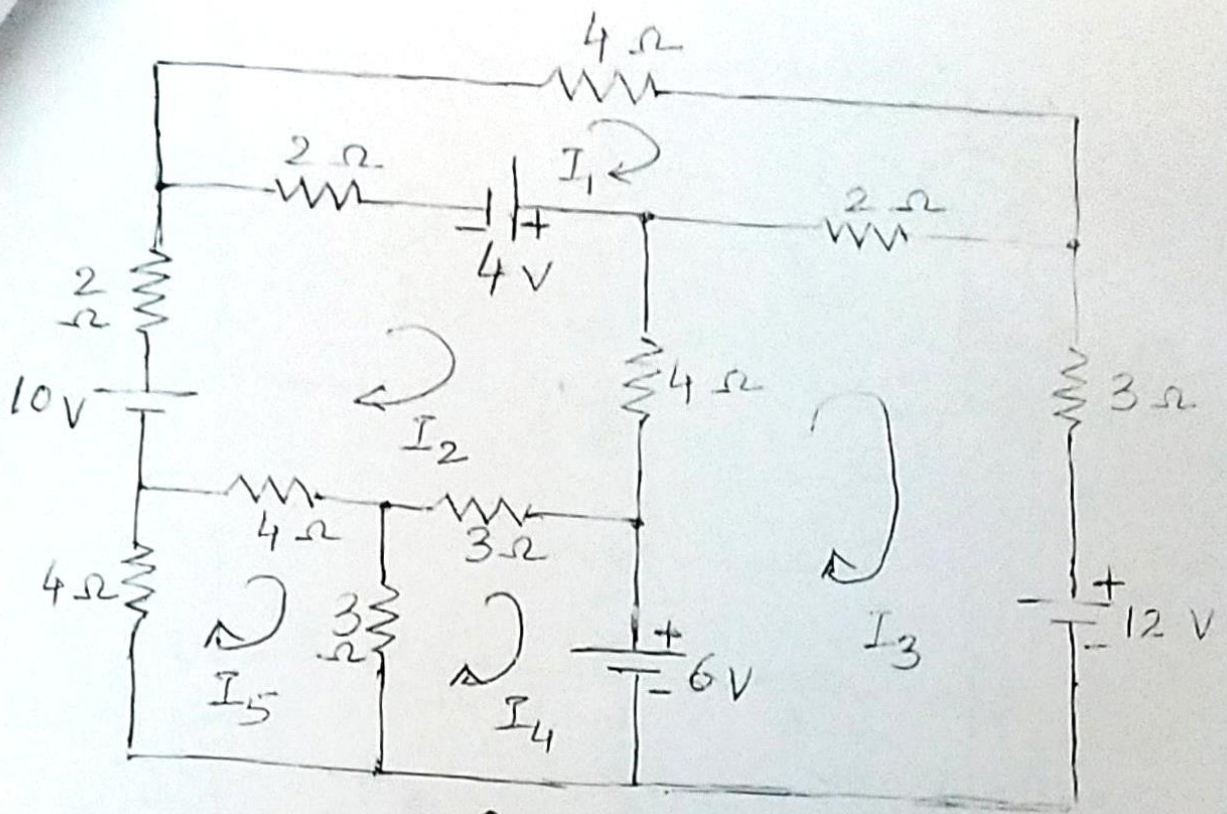
PROBLEMS

- Q1. Find R and PF of the circuit.(FIG.)
 Ans. $[1.21\Omega, 0.638 \text{ lap}]$
- Q2. The circuit carries a current of 35A. Find R, r, L and frequency of the applied voltage and its magnitude. (FIG.)
 Ans. $[0.714\Omega, 0.517\Omega, 0.073 \text{ mH}, 2.475 \text{ kHz}, 31 \text{ V}]$
- Q3. Given $R = 2\Omega$, $L = 0.3 \text{ H}$, $i_R = 10\sqrt{2} \cos(10t + 45^\circ) \text{ A}$ with V_i as reference. Draw the phasor diagram showing $\bar{V}_C$, $\bar{V}$, $\bar{I}_L$. Determine therefore (i) C, (ii) $\bar{I}_L$ and $\bar{I}_L(t)$, (iii) $\bar{V}$ and $v(t)$. (FIG.)
 Ans. $[0.05 \text{ F}; 9.43 \text{ A}, 9.43 \cos(10t - 90^\circ) \text{ A or } 9.43 \sin 10t \text{ A}; 28.28\angle 0^\circ \text{ V}, 28.28 \cos 10t \text{ V}]$
- Q4. Determine all mesh currents, determine voltages V_1 & V_2 . (FIG.)
- Q5. Obtain the loop current matrix. (FIG.)
- Q6. Obtain the loop currents. (FIG.)
 Ans. $[I_1 = 20/17 \text{ A}, I_2 = 10/17 \text{ A}]$
- Q7. Find current through 6Ω resistor. (FIG.)
 Ans. $[2.5 \text{ A}]$
- Q8. Find current through 6Ω resistor. (FIG.)
 Ans. $[-5 \text{ A}]$
- Q9. Determine node voltages. (FIG.)
 Ans. $[V_1 = 3.2 \text{ V}; V_2 = 1.6 \text{ V}; V_3 = -1.6 \text{ V}]$
- Q10. Obtain the node voltage matrix, by inspection. (FIG.)
 Ans.
$$\begin{bmatrix} 0.3 & -0.2 & 0 & 0 \\ -0.2 & 1.325 & -0.125 & -1 \\ 0 & -0.125 & 0.5 & -0.125 \\ 0 & -1 & -0.125 & 1.625 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -3 \\ 0 \\ 6 \end{bmatrix}$$
- Q11. Find I_o using nodal analysis. (FIG.)
 Ans. $[I_o = 0.3125 \text{ mA}]$
- Q12. Find a two-element parallel network with the same impedance. (FIG.)
- Q13. Find a two-element series network with the same impedance. (FIG.)
- Q14. Find V_X using node voltage method. (FIG.)
- Q15. Calculate current through a $100\mu\text{F}$ capacitor which is subjected with a voltage shown in the fig., draw the current waveform.
- Q16. Under resonance condition, Find (i) inductive reactance, (ii) current I, (iii) voltage drop across resistor, inductor & capacitor, (iv) Q- factor of the circuit, (v) value of L & C if $f = 5.5 \text{ kHz}$. (FIG.)
- Q17. Design a series resonant circuit which has a lower half power frequency of 2000 Hz and upper half power frequency of 2300 Hz. The inductance of non-ideal inductor, having some internal resistance is 80 mH. When the inductor is connected to a 10 V (rms), 100Hz source the current through the circuit is $0.15\angle -81^\circ \text{ A}$. (i) what will be the value of C so that the circuit will be at resonance? (ii) Determine Q-factor of the circuit at resonance. (FIG.)
- Q18. Find current through 2Ω resistor using source transformation. (FIG.)
 Ans. $[1.0909 \text{ A}]$
- Q19. Find current through 4Ω resistor. (FIG.)
 Ans. $[50 \text{ A}]$
- Q20. Using supermesh analysis, find current through 4Ω resistor. (FIG.)
 Ans. $[2 \text{ A}]$
- Q21. Find the voltage across the dependent voltage source. (FIG.)

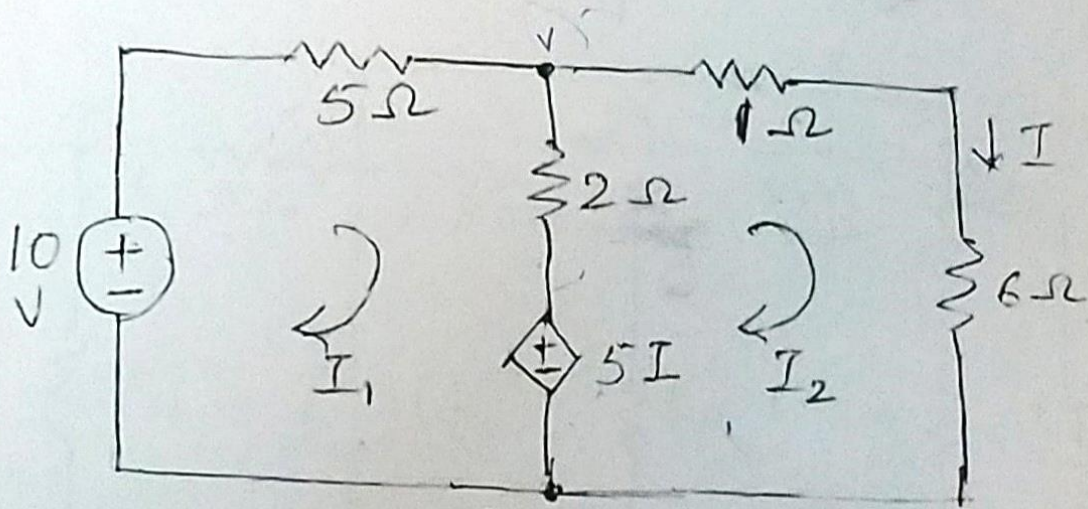
- Q22. Find current through the dependent current source. (FIG.)
- Q23. Find the voltage across the dependent current source. (FIG.)
- Q24. Find equivalent resistance between A-B. (FIG.)
Ans. $[9.8919 \Omega]$
- Q25. Find equivalent resistance between point a-b. (FIG.)
Ans. $[8 R/7]$
- Q26. A source of $e(t) = 20 e^{-4t}$ is applied to a series circuit of resistance 2Ω and inductance 3 H . Find voltage drop across resistance and inductance.
Ans. $[-4 e^{-4t} \text{ (V)}; 24 e^{-4t} \text{ (V)}]$
- Q27. Calculate rms and average value of the waveform, over a complete cycle. (FIG.)
Ans. $[\sqrt{8/3} \text{ V}; 0 \text{ V}]$
- Q28. Determine the rms value of the waveform. (FIG.)
Ans. $[5.77 \text{ A}]$
- Q29. Using phasors add the following sinusoids
 $V_1 = 20 \cos(\omega t - 45^\circ)$
 $V_2 = 10 \sin(\omega t + 30^\circ)$
Reduce the sum $V_s(t) = V_1(t) + V_2(t)$ to a single term.
Ans. $[V_s(t) = 29.77 \cos(\omega t - 50^\circ)]$
- Q30. A 50Hz sinusoidal voltage $V = 141.4 \sin \omega t$ is applied to a series R-L circuit with $R = 3\Omega$ and $L = 0.0127 \text{ H}$. Calculate (i) the effective value of steady-state current and its angle, (ii) average power dissipated in the circuit, (iii) power factor of the circuit.
Ans. $[20\angle -53.1^\circ \text{ A}; 1200 \text{ W}; 0.6 \text{ (lagging)}]$



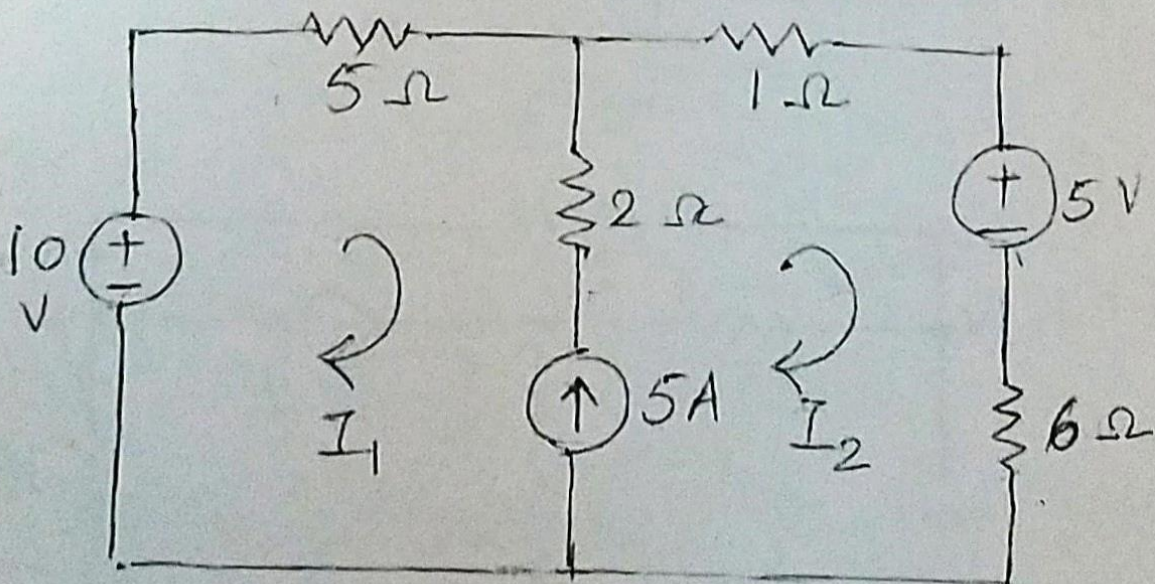




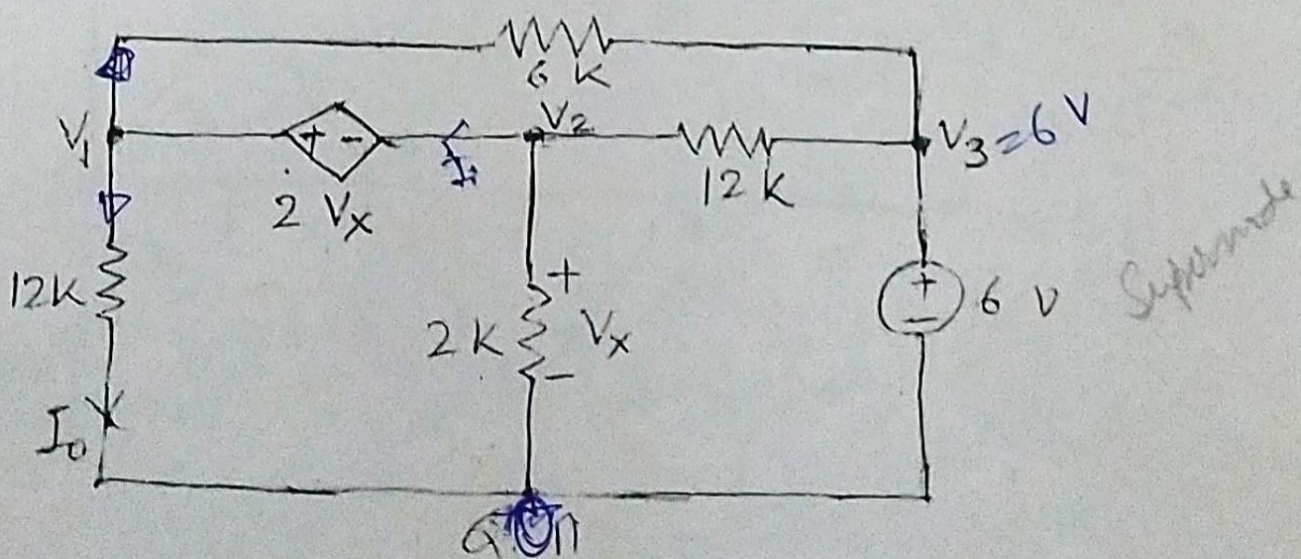
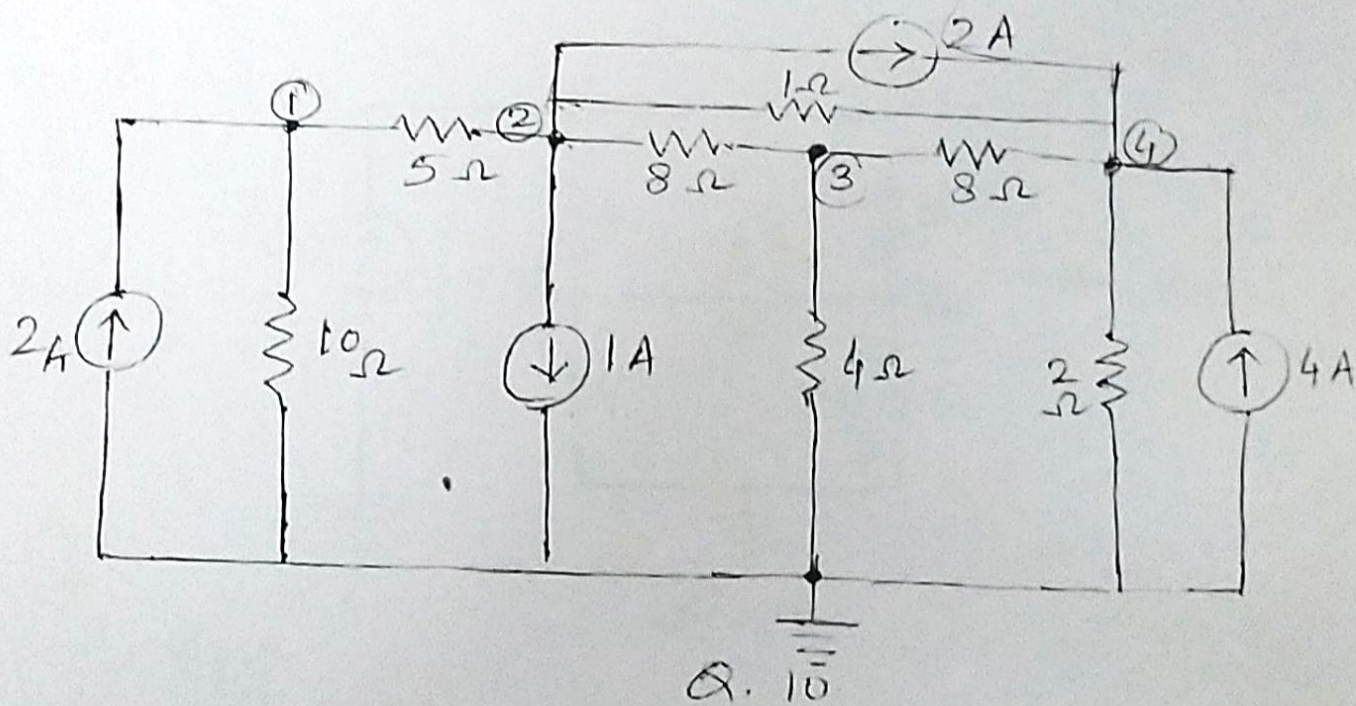
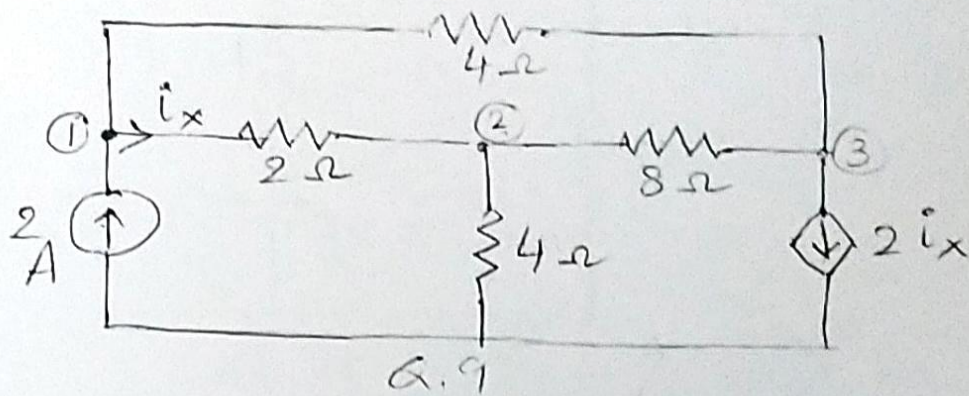
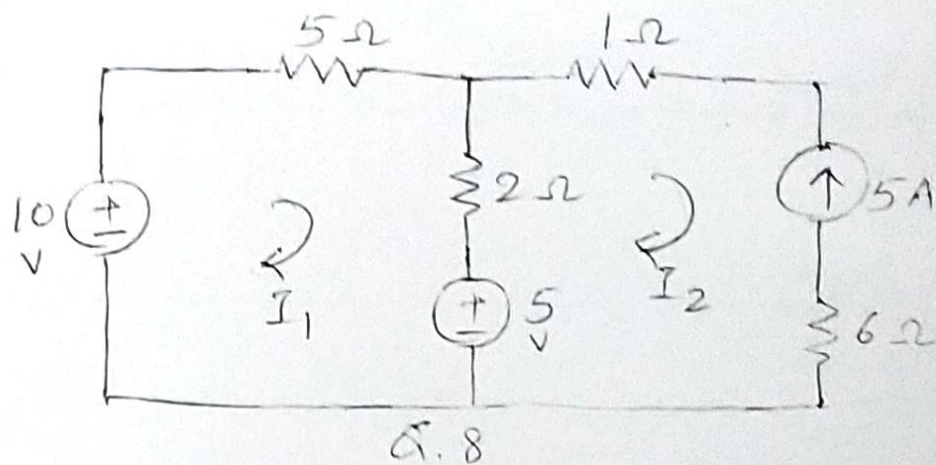
Q.5

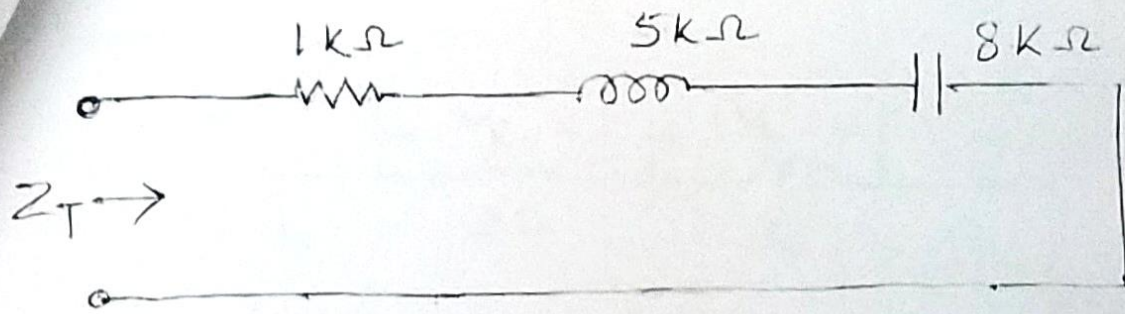


Q.6

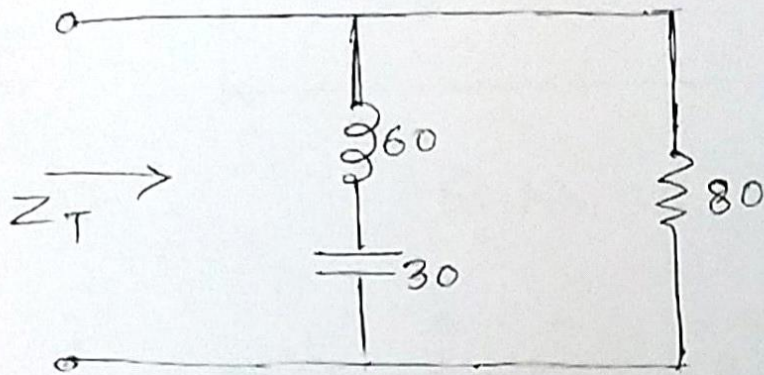


Q.7

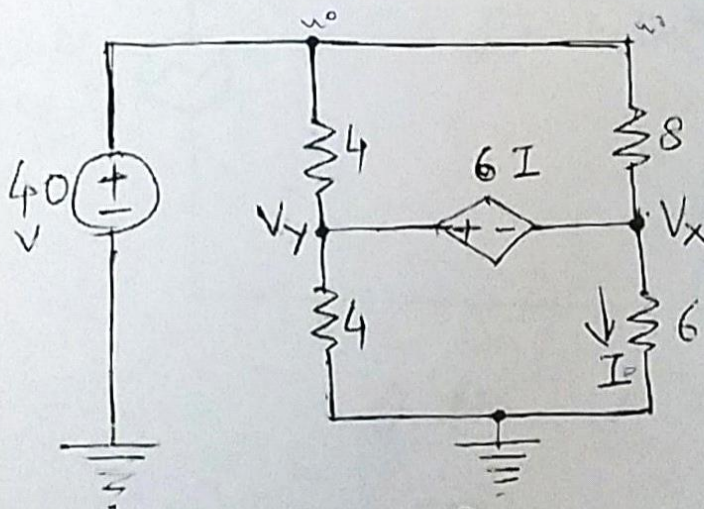




Q. 12

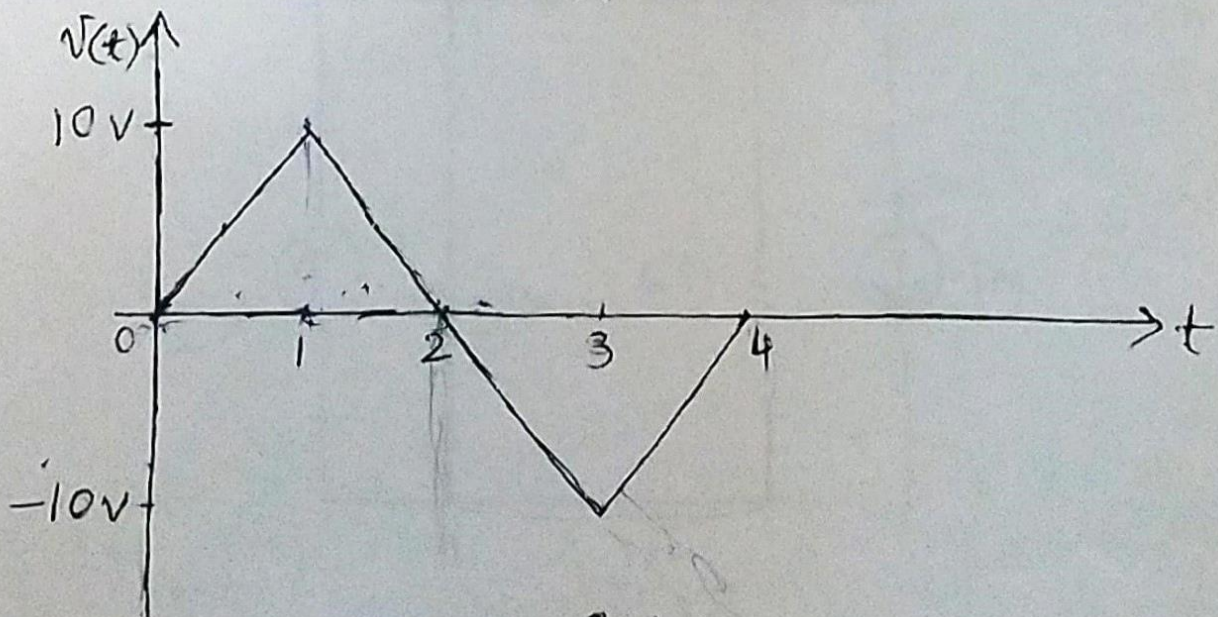


Q. 13

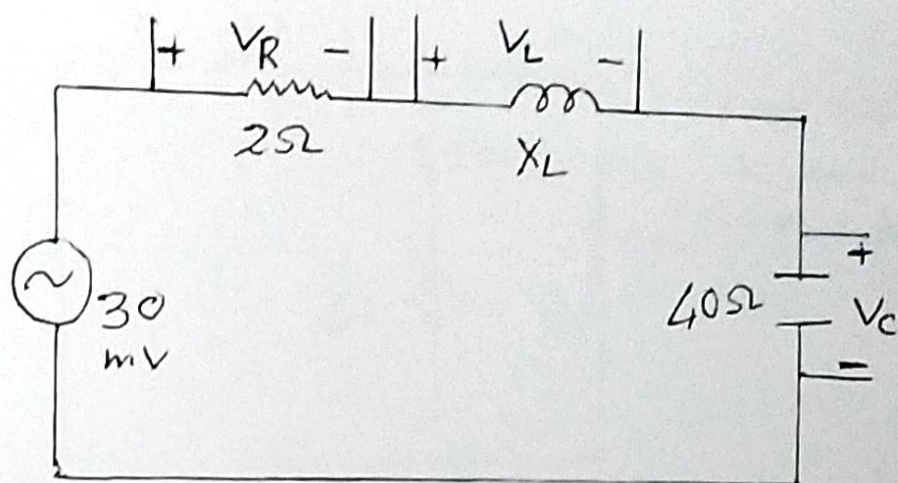


SuperNode

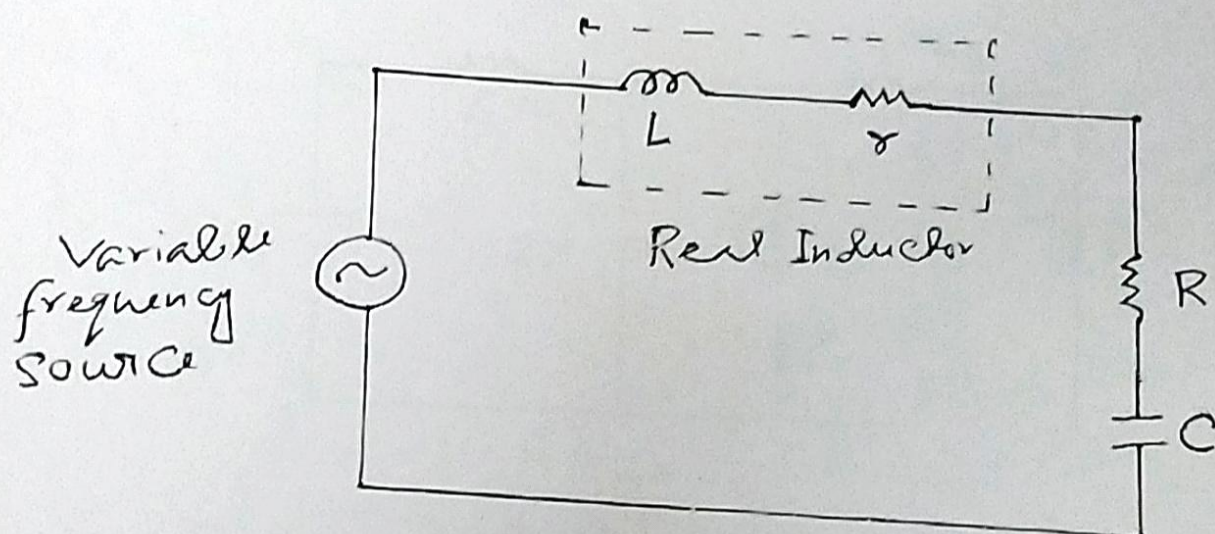
Q. 14



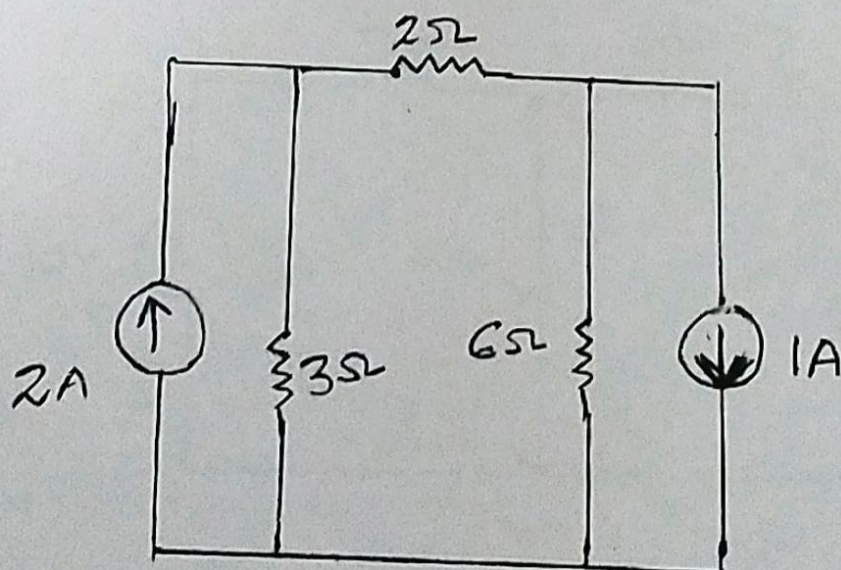
Q. 15



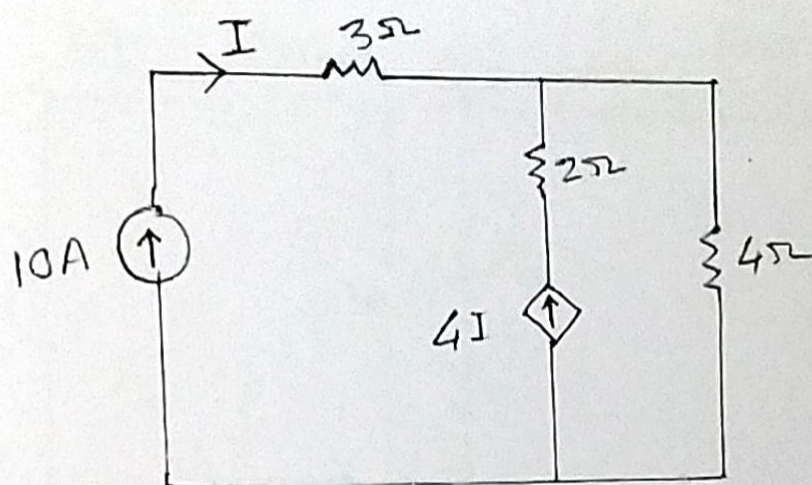
Q. No. 16



Q. No. 17

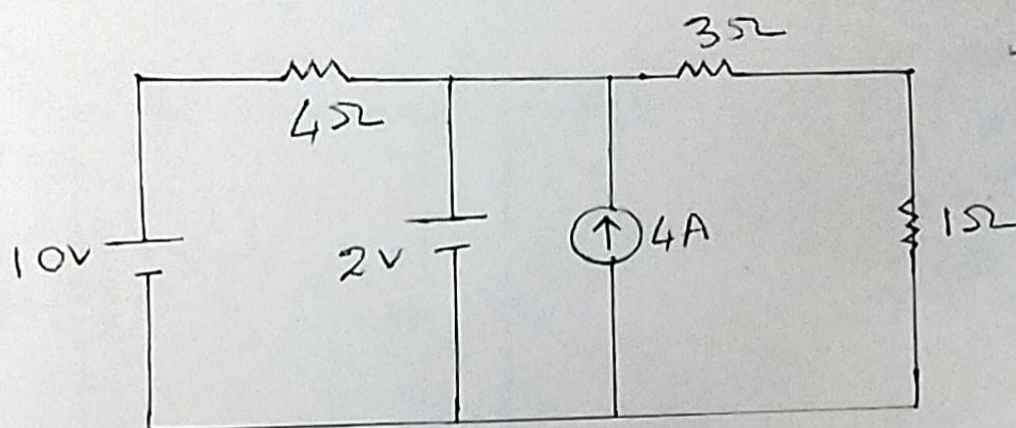


Q. No. 18

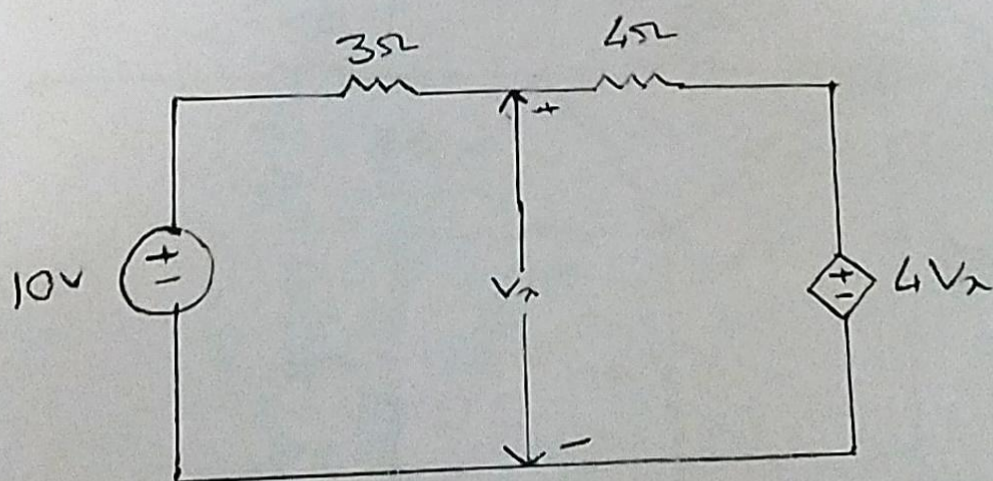


Find current
through 4Ω
resistor.

Q. No. 19

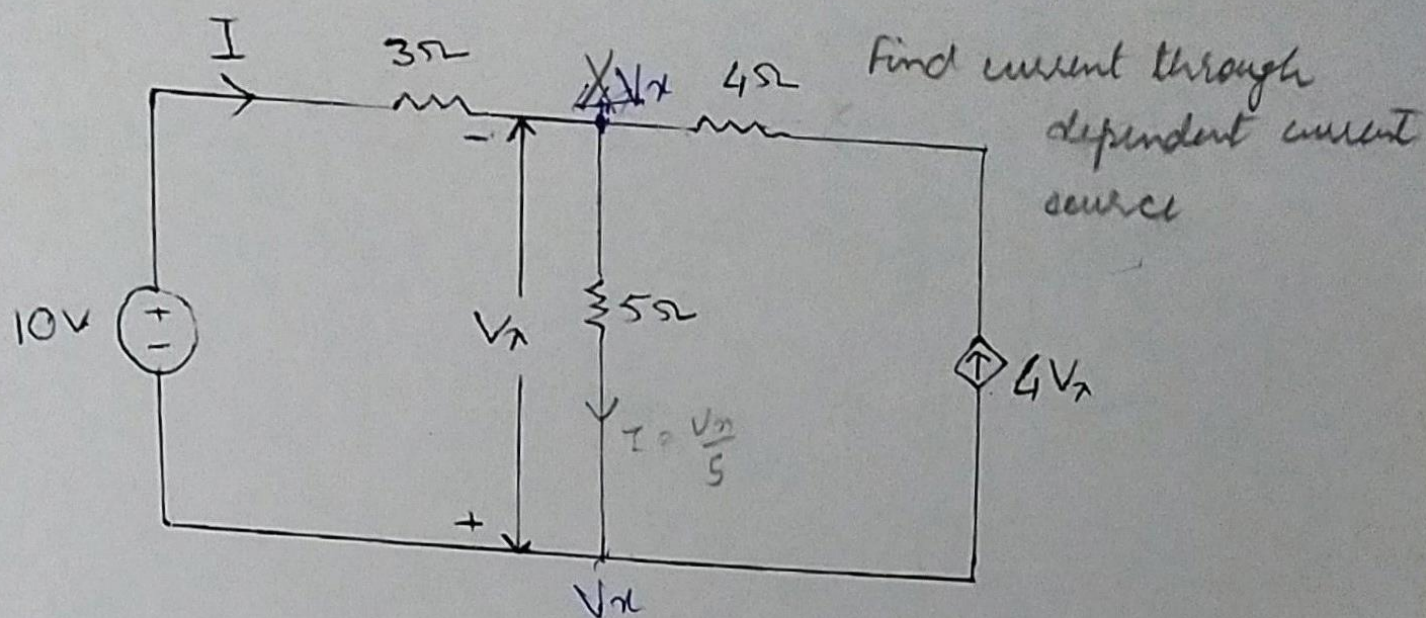


Q. No. 20

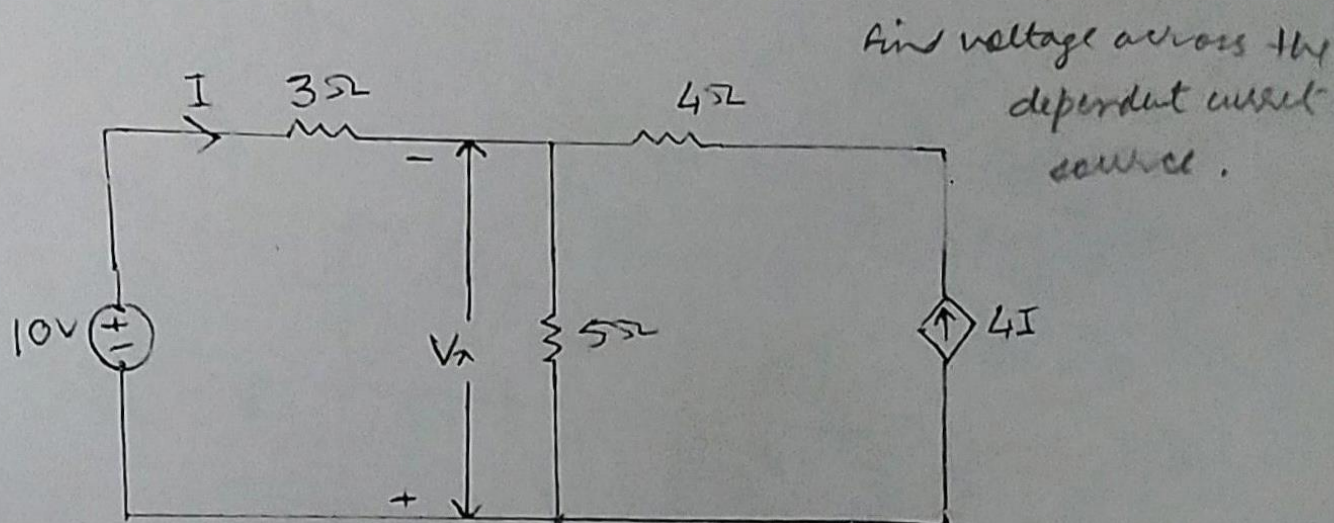


Q. No. 21

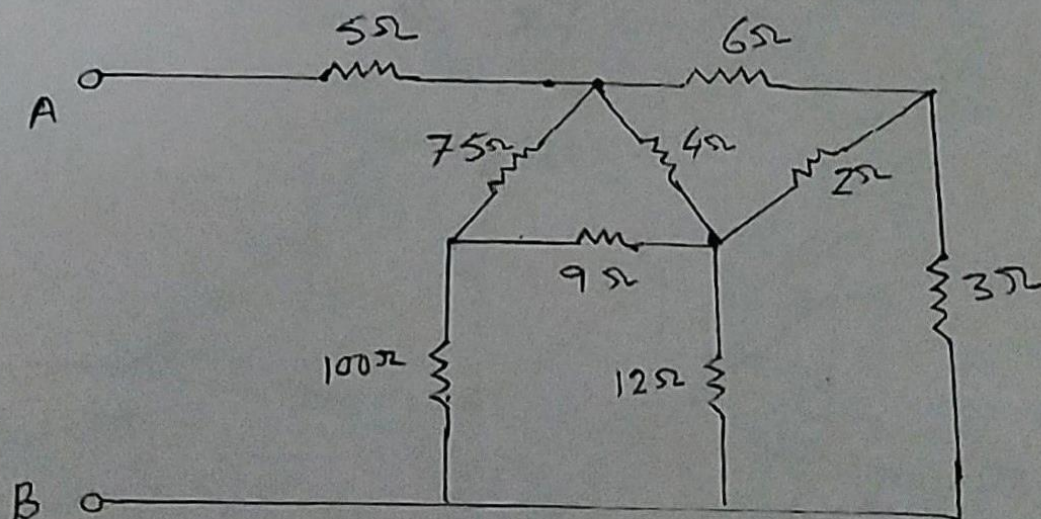
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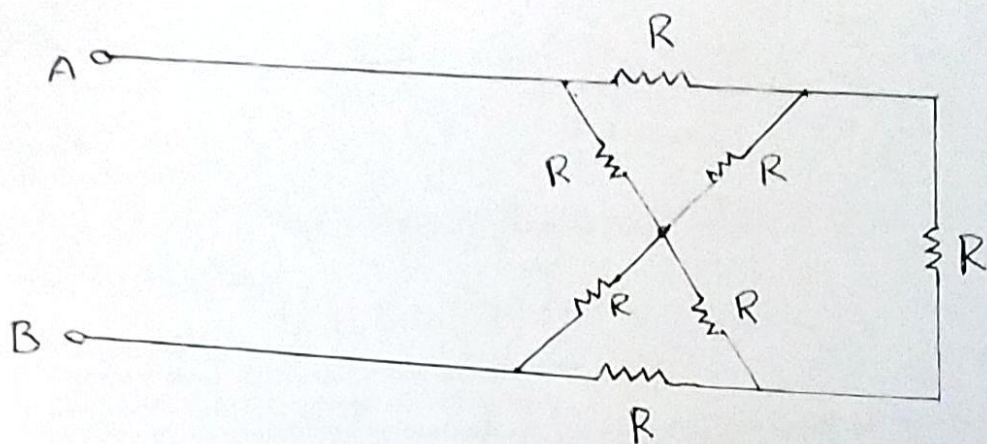
Q No. 22



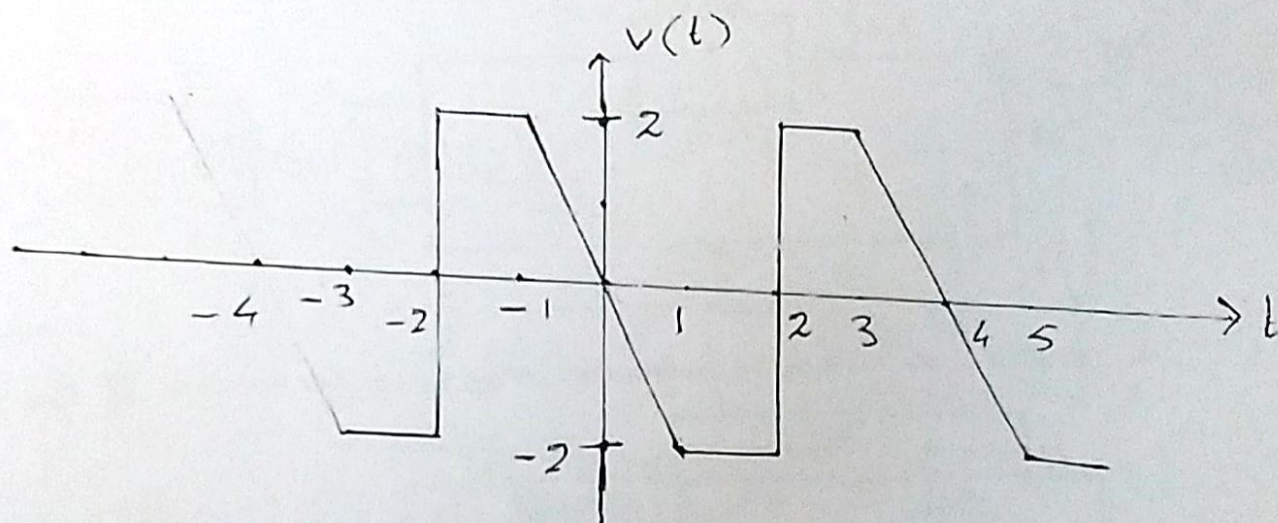
Q No. 23



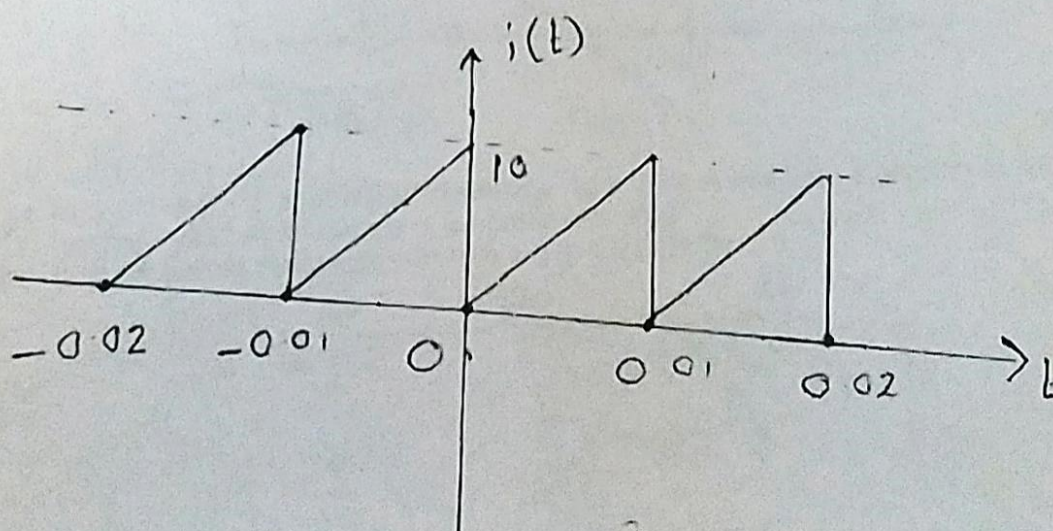
Q No. 24



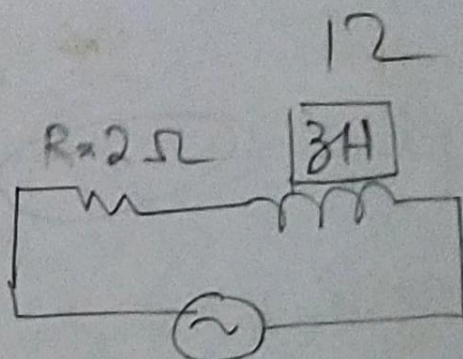
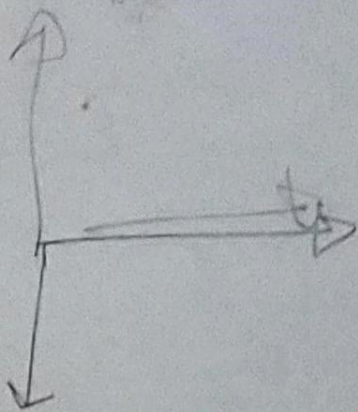
Q. No 25



Q. No 27



Q. No 28



$$v(t) = 20 e^{-4t}$$

1) V drop across R and L $\cdot 20 e^{-4t}$

$$= 20 (\cos(4t) - \sin(4t))$$

$$= 20 \cos(4t) - 20 \sin(4t)$$

$$\begin{matrix} -4e^{-4t} \\ 24e^{-4t} \end{matrix}$$

$$\sqrt{2^2 + 12^2}$$

$$\sqrt{148}$$

$$-4e^{-4t}$$

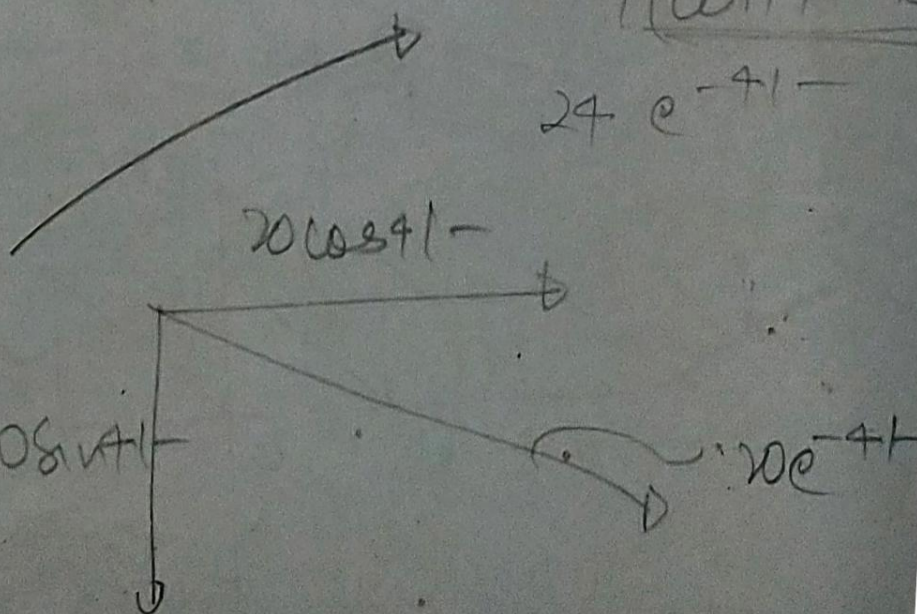
$$4(\omega + j - i)$$

$$24 e^{-4t}$$

$$20 \cos(4t)$$

$$-20 \sin(4t)$$

$$20 e^{-4t}$$



BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BE
BRANCH: ALL

SEMESTER : I
SESSION : MO/15

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME: 1.5 HOURS

FULL MARKS: 25

INSTRUCTIONS:

1. The total value of the questions are 30 marks.
2. Candidates may attempt for all 30 marks.
3. In those cases where the marks obtain exceed 25 marks. The excess will be ignored.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. The missing data, If any, may be assumed suitably.

Q1. (a) Write notes on Supernode analysis.

[2]

(b) Find R and power factor of the circuit in fig. (1).

[3]

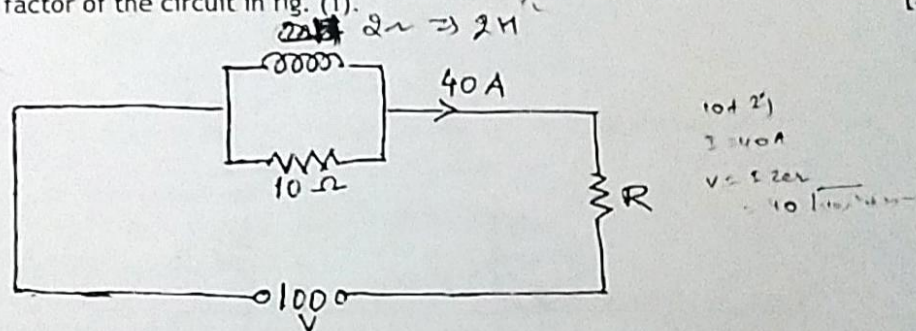


Figure 1

Q2. Obtain node voltage matrix, by inspection, for the circuit shown in fig. (2).

[5]

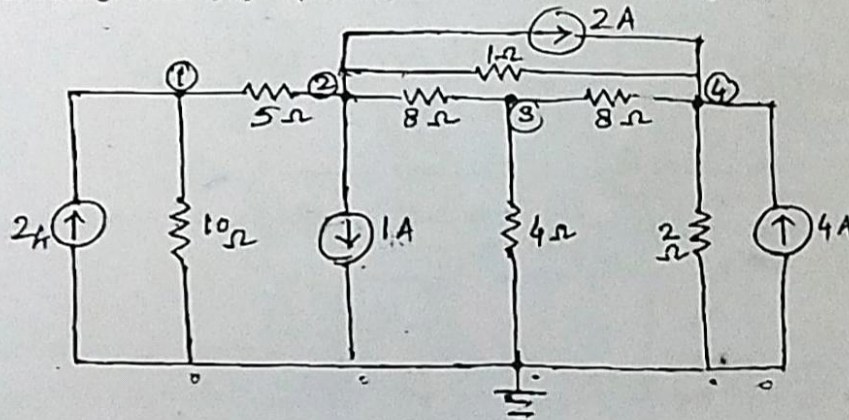


Figure 2

Q3. (a) In a network, it is possible to identify four meshes and four independent nodes, which method of circuit analysis is to be preferred? Why?

[1]

(b) Find equivalent resistance between a-b as shown in fig. (3).

[4]

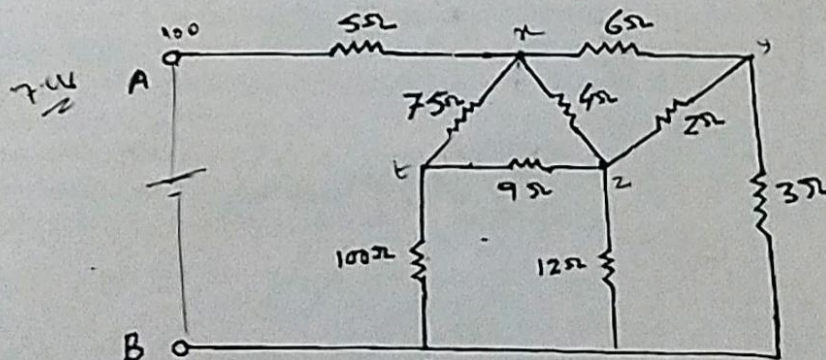


Figure 3

- Q4. Two coil A and B are connected in series across 250V, 50Hz supply. The resistance of A is 5Ω and the inductance of B is 0.015 H . If the input from the supply is 4 kW and 3 kVAR , find the inductance of A and resistance of B. Also calculate voltage across each coil. [5]
- Q5. (a) Derive an expression of resonant frequency for a series RLC circuit. Show graphically variations of various parameters versus frequency both under series and parallel resonance. [3]
- (b) Define and explain the terms: admittance, susceptance and conductance. [2]
- Q6. (a) Find current through 2Ω resistor using source transformation (fig. 4). [2]

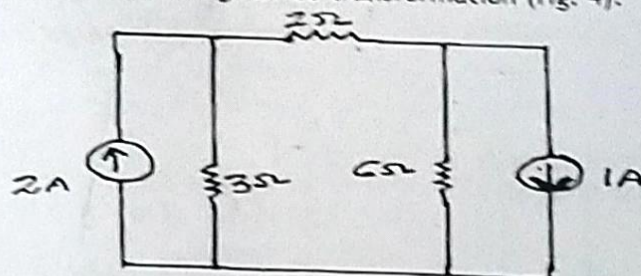


Figure 4

- (b) Find current through 6Ω resistor shown in circuit of fig. (5). [3]

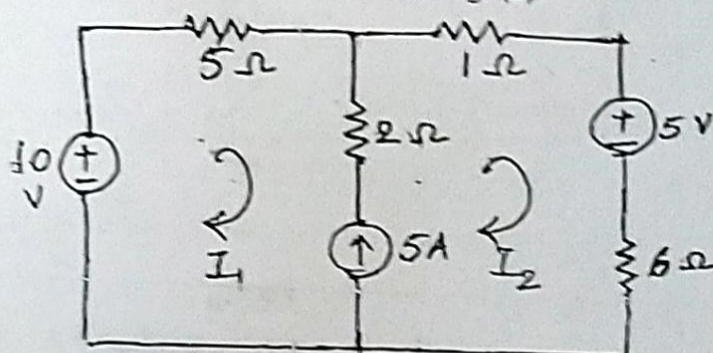


Figure 5

***** 01.10.2015 M*****

$$(9 \times 4 + 2) =$$

$$(9 \times 4 + 2) =$$

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BE
BRANCH: ALL

SEMESTER: I
SESSION: MO/16

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME: 1.5 HOURS

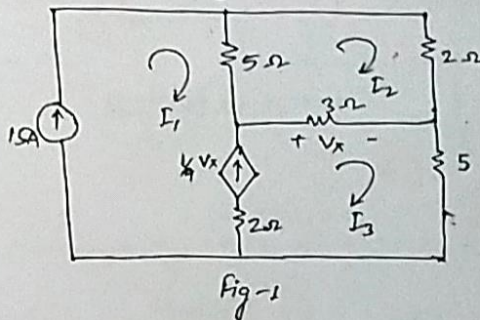
FULL MARKS: 25

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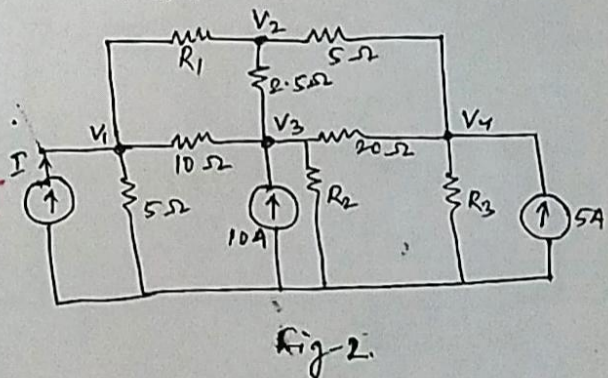
- Q1 (a) What do you understand by (i) active and passive elements (ii) unidirectional and bidirectional elements? [2]
(b) Find the currents in the circuit shown in Fig. 1. [3]

$I_1 = 1.5 \text{ A}$
 $I_2 = 12.4 \text{ A}$
 $I_3 = 16.3 \text{ A}$



- Q2 (a) What are Kirchoff's laws? Explain. [2]
(b) The node voltages in the circuit in Fig 2. are as follows: $V_1=10\text{V}$, $V_2=20\text{V}$, $V_3=30\text{V}$ and $V_4=10\text{V}$. Find the values of R_1 , R_2 , R_3 and i . [3]

$R_1 = 5$
 $i = -2$
 $R_2 = 10$
 $R_3 = \frac{5}{4} = 1.25$



- Q3 (a) Define form factor and peak factor. [2]
(b) Show that power consumed in a pure resistive circuit is constant but it is fluctuating. [3]

- Q4 (a) Draw the phasor diagram of the following voltages: [2]

$v_1 = 100 \sin 100 t$
 $v_2 = 120 \sin (100 t - \pi / 3)$
 $v_3 = 50 \sin (100 t + \pi / 4)$
 $v_4 = 200 \cos 100 t$

Also find the resultant voltage and its rms value.

- (b) A coil is connected in parallel with a $200\mu\text{F}$ capacitor across a 200V , 50Hz supply. The coil takes a current of 8A and loss in the coil is 960W . Calculate the (i) resistance of the coil (ii) inductance of the coil (iii) power factor of the circuit. Draw the power triangle. [3]

$R = 15 \Omega$, $L = 0.063 \text{ H}$
 $Z_{\text{total}} = 25.77 \angle 51.8^\circ$
 $\text{pf} = 0.61$ lagging leading -

PTO

- Q5 (a) Explain the use of capacitor in power factor improvement. [2]
 (b) An inductive coil of $(6+j8)\Omega$ impedance is connected to 100V, 50Hz AC supply. It is desired [3]
 to improve the power factor of supply current to 0.8 lagging by connecting a capacitor in
 parallel. Determine the value of capacitance.

- Q6 (a) Explain how different parameters vary with frequency in a series resonant circuit. [2]
 (b) A 10mH coil is connected in series with a loss-less capacitor to a variable frequency source [3]
 which supplies a constant voltage of 10V. The circuit current has a maximum value of 0.1 A
 at a frequency of 80Hz. Calculate

- a. The capacitance of capacitor $396\mu F$
 b. Q factor of the coil 1000 0.05
 c. Half-power frequencies
 d. Bandwidth of the circuit

$$R = 100\Omega$$

$$1000 = \frac{R}{Z} = \frac{100}{10 \times 10^{-3}} = \frac{100 \times 10^3}{10} = 10000$$

*****25.10.16*****M

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BE
BRANCH: ALL

SEMESTER : I
SESSION : MO/15

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME: 1.5 HOURS

FULL MARKS: 25

INSTRUCTIONS:

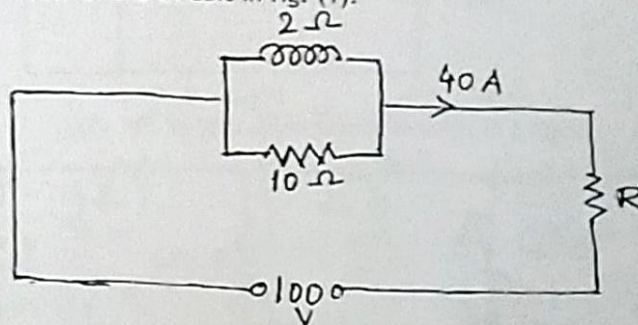
1. The total value of the questions are 30 marks.
2. Candidates may attempt for all 30 marks.
3. In those cases where the marks obtain exceed 25 marks. The excess will be ignored.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. The missing data, If any, may be assumed suitably.

Q1. (a) Write notes on Supernode analysis.

[2]

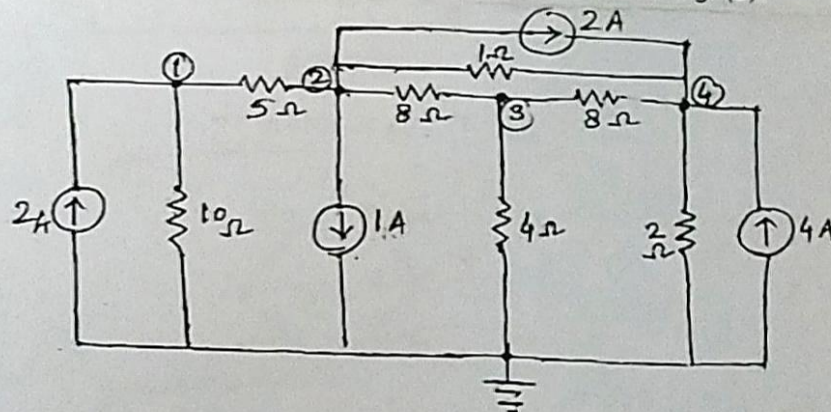
(b) Find R and power factor of the circuit in fig. (1).

[3]



Q2. Obtain node voltage matrix. by inspection, for the circuit shown in fig. (2).

[5]

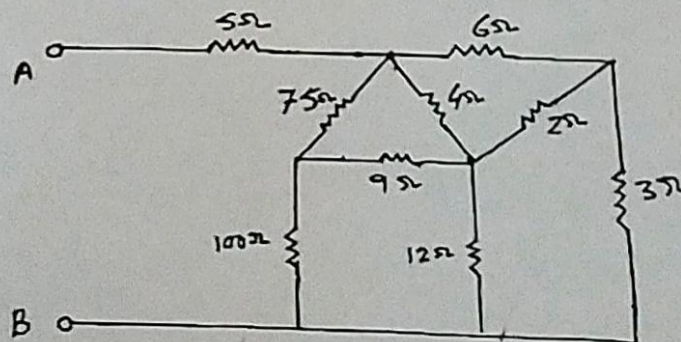


Q3. (a) In a network, it is possible to identify four meshes and four independent nodes, which method of circuit analysis is to be preferred? Why?

[1]

(b) Find equivalent resistance between a-b as shown in fig. (3).

[4]



- Q4. Two coil A and B are connected in series across 250V, 50Hz supply. The resistance of A is 5Ω and the inductance of B is 0.015 H . If the input from the supply is 4 kW and 3 kVAR , find the inductance of A and resistance of B. Also calculate voltage across each coil. [5]
- Q5. (a) Derive an expression of resonant frequency for a series RLC circuit. Show graphically variations of various parameters versus frequency both under series and parallel resonance. [3]
 (b) Define and explain the terms: admittance, susceptance and conductance. [2]
- Q6. (a) Find current through 2Ω resistor using source transformation (fig. 4). [2]

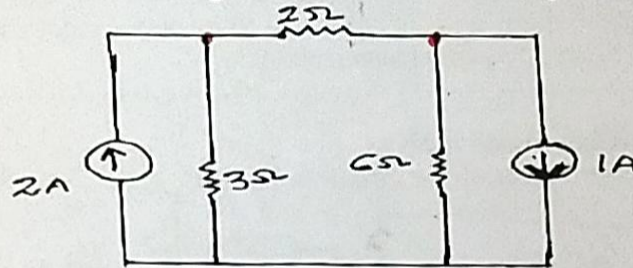


Figure 4

- (b) Find current through 6Ω resistor shown in circuit of fig. (5). [3]

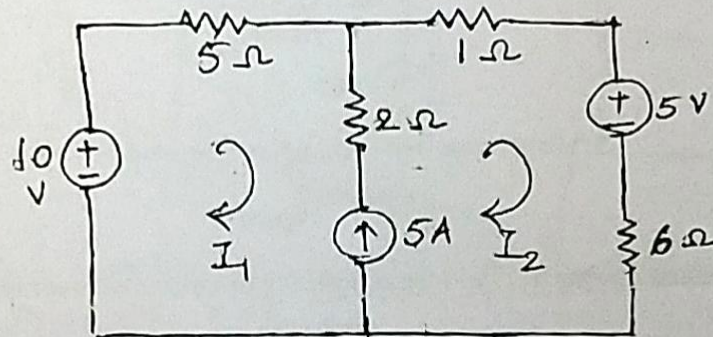


Figure 5

***** 01.10.2015 M*****

1. What is source conversion technique. Find the current I using source conversion technique in fig1 (3)
2. Write down Mesh equations to find the different currents in the circuit fig2 (2)
3. Write down Nodal equations to find the different currents in the circuit fig3 (2)
4. Find the equivalent resistance between terminals A and B fig.4 (3)

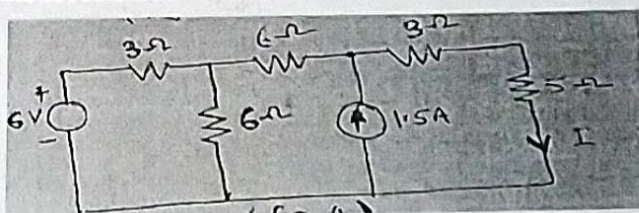


Figure 1

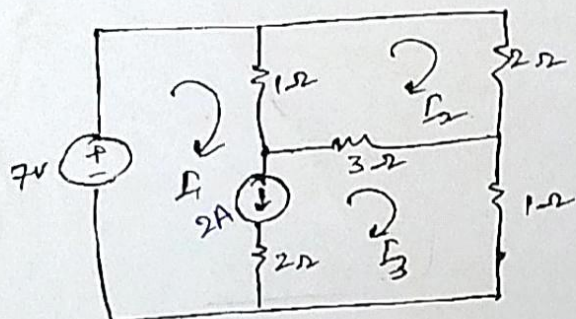


Figure 2

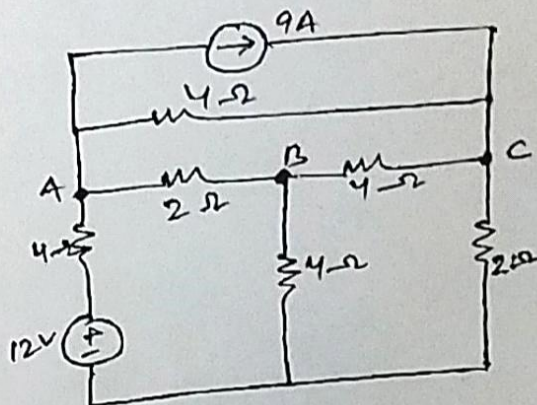


Figure 3

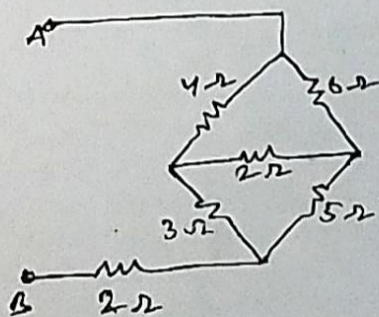


Figure 4

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS : B.E.
BRANCH : ALL

SUBJECT: EE 2101 BASIC ELECTRICAL ENGINEERING

SEMESTER : II
SESSION : SP/2011

FULL MARKS : 60

TIME : 2 HOURS

INSTRUCTIONS:

1. The question paper contains 7 questions each of 12 marks and total 84 marks.
2. Candidates may attempt any 5 questions maximum of 60 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.

- Q.1(a) Consider a 230V, 100W incandescent lamp. Determine the lamp resistance and current. [2]
(b) State the principle of superposition. [2]
(c) In circuit of Fig.1 $V_0 = 16V$, find I_s . [4]

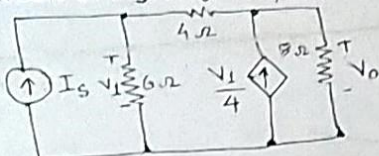


Fig.1

- (d) Determine V_x in the circuit of Fig.2 [4]

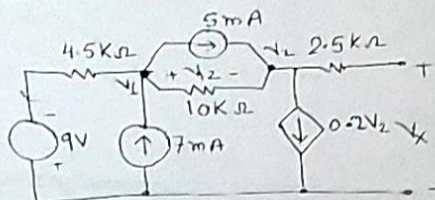


Fig.2

- Q.2(a) Explain how a practical current source is converted to a voltage source [2]
(b) Prove the maximum power transfer theorem. [4]
(c) A bulb rated 110V, 60W is connected with another bulb in series rated 110V, 100W across a 220V mains. Calculate the resistance which should be joined so that both bulb may take their rated power. [4]
(d) Define ideal and practical voltage source. [2]

- Q.3(a) Find the effective value of a voltage $v = 150 + 50\sin\omega t + 25\cos\omega t$. [2]
(b) A voltage of $141.4\sin 1256t$ volt is applied across a 100 ohm resistor. Find the current in the resistor if it is fed through a half wave rectifier. [2]
(c) A non-inductive resistance is connected in series with a coil and a capacitor. The circuit is connected to a single - phase a.c. supply. If the voltage drop across resistance is 25V, drop across coil is 40V, drop across capacitor is 55V and drop across resistance and coil together is 50V when current flowing through the circuit is 0.345A. Find the applied voltage and the power loss in the coil. [4+4]

- Q.4(a) Two impedances, Z_1 and Z_2 are connected in parallel. The first branch takes a leading current of 16A and has a resistance of 5Ω , while the second branch takes a lagging current at power factor 0.8. The total power supplied is 5 KW, the applied voltage being $100 + j 200$ V. Determine the complex expressions for the branch and total currents, and the circuit constants. [4+4]
(b) Define and explain RMS and Average value of a sinusoid voltage. [2+2]

- Q.5(a) Explain the necessary and advantage of three phase AC. [2]
(b) Explain power factor measurement by means of two wattmeters in a three phase ac circuit. [4]
(c) Explain what is meant by phase sequence in 3-phase system. [2]
(d) A 3 phase 400V circuit supplies as delta connected load having phase impedances of $Z_{AB} = 25/_0^\circ$, $Z_{BC} = 25/_30^\circ$, and $Z_{CA} = 25/_-30^\circ$. Two wattmeters are connected in the circuit to measure the load power. Determine the wattmeter readings if their current coils are in lines A and B. [4]

- Q.6(a) Explain dot convention for coupled coils [2]
(b) What do you mean by eddy current. [2]
(c) An iron ring has a cross section of 3 sq cm and a mean diameter of 25cm. An air gap of 0.4cm has been cut across the section of the ring. The ring is wound with a coil of 200 turns through which a current of 2A is passed. If the total magnetic flux is 0.24 mWb, find the relative permeability of iron, assuming no magnetic leakage. [4]
(d) Define reluctance and mmf. [4]
- Q.7(a) - Explain controlling and damping torque.. [2+2]
(b) A moving coil instrument has a resistance of 10 ohm and gives full scale deflection when carrying a current of 50mA. Show how it can be adopted to measure current upto 1000A [4]
(c) Write short on working of moving Iron Instrument. [4]

29.4.2011M

CLASS: BE
BRANCH: ALL

SEMESTER : SHORT
SESSION : SS/18

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME: 03:00 Hours

FULL MARKS: 100

INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt any 5 questions maximum of 100 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

- Q.1(a) Differentiate between unidirectional and bidirectional elements. Mention one example of each. [4]
(b) Describe Kirchoff's laws. [6]
(c) By nodal method, find the current through 2V source for the circuit shown in Fig. 1. [10]

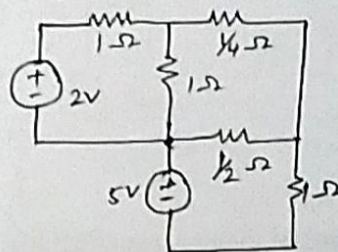


Fig-1

- Q.2(a) Define phase and phase difference? Explain with the help of phasor and wave diagrams. [4]
(b) Derive the relation between RMS and average value of a sinusoidal quantity. [6]
(c) An inductive load is connected in series with a non-inductive resistance of 8Ω . The combination is connected across an AC supply of 100V, 50Hz. A voltmeter connected across the non-inductive resistor and then across the inductive load gives the reading of 64V and 48V respectively. Calculate the following: (i) impedance of the load (ii) total impedance (iii) total power taken from supply (iv) power factor of the load (v) power factor of the whole circuit. Also draw the phasor diagram. [10]
- Q.3(a) Explain how power factor in RC series circuit is different from RL series circuit. [4]
(b) A coil of resistance 40Ω and inductance $0.75H$ forms part of a series circuit for which the resonant frequency is 55Hz. If the supply is 250V, 50Hz, find (i) line current (ii) power factor (iii) voltage across the coil. [6]
(c) In the circuit shown in Fig.2, determine what 50Hz voltage applied across AB will cause a current of 10A to flow in the capacitor. [10]

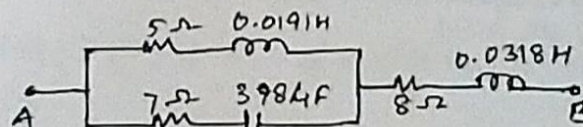


FIG.2

- Q.4(a) State and explain the Thevenin's theorem. [4]
(b) State and prove the maximum power transfer theorem for AC circuits. [6]
(c) Find Norton's equivalent across terminals X-Y for the circuit given in Fig. 3. [10]

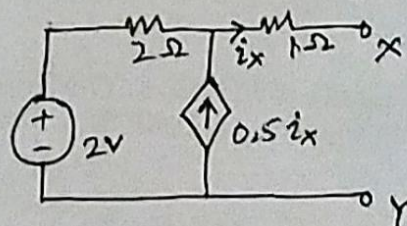


Fig-3

- Q.5(a) Define and explain phase sequence. [4]
 (b) Derive the expression of power consumed by a balanced three phase star system. [6]
 (c) Explain the two-wattmeter method for measurement of three-phase power. Also draw the phasor diagram. [10]

- Q.6(a) Define (i) magnetic field strength (ii) MMF [4]
 (b) What is hysteresis loss? How it can be minimized in a transformer core? Explain. [6]
 (c) A magnetic core made of annealed sheet steel has the dimensions as shown in Fig. 4. The cross section is 25cm^2 everywhere. The flux in branches A and B is $3500\mu\text{Wb}$ but that in the branch c is zero. Find the required AT for coil A and C. relative permeability of sheet steel is 1000. [10]

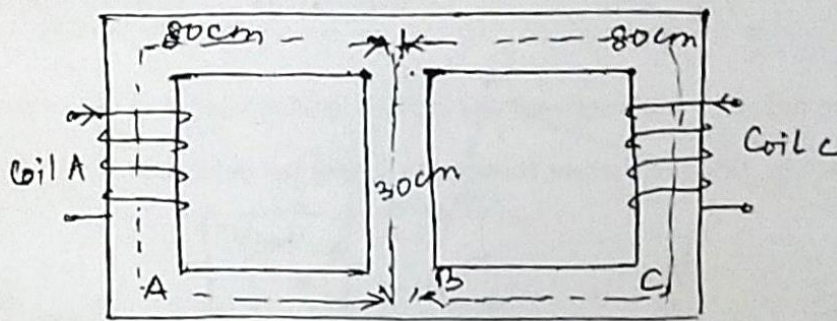


Fig-4

- Q.7(a) List advantages and disadvantages of moving iron instruments. [4]
 (b) Explain attraction and repulsion types of moving iron instruments. [6]
 (c) Obtain the expression for input impedance of the circuit shown in Fig. 5. [10]

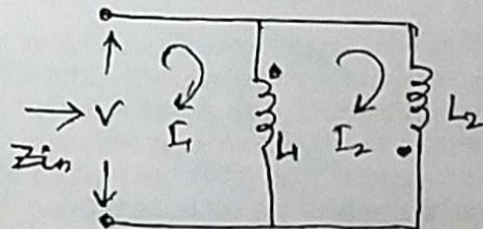


Fig-5

*****14/06/18*****

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(SHORT SEMESTER EXAMINATION)

CLASS : BE
BRANCH : ALL

SEMESTER : SS/2013
SESSION : 2012-13

SUBJECT : EE 2101 BASIC ELECTRICAL ENGINEERING

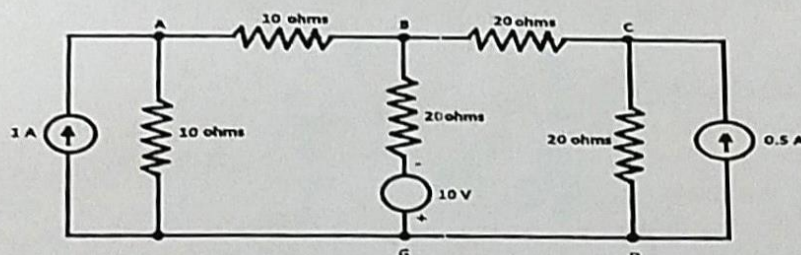
TIME : 3 HOURS

FULL MARKS : 100

INSTRUCTION :

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
 2. Candidates may attempt for all questions.
 3. In those cases where the marks obtained exceed 100 marks, the excess will be ignored.
 4. The missing data, if any, may be assumed suitably.
 5. Before attempting the question paper, be sure that you have got the correct question paper.
 6. Tables/Data hand book/Graph Paper etc. to be supplied to the candidates in the examination hall.
-

- Q1. (a) Define Impedance, Reactance, Susceptance and Admittance. [4]
(b) What are the different types of current and voltage source? Give example of each. [6]
(c) Using nodal analysis, determine current in each branch of the network shown. also find the total power loss in the network.



- Q2. (a) Explain the terms - In phase, out of phase, form factor and peak factor. [4]
(b) Two currents $i_1 = 7 \sin \omega t$ and $i_2 = 10 \sin(\omega t + \pi/3)$ are fed to a common conductor. What is the resultant current and its rms value. [6]
(c) A coil is connected in series with a non-inductive resistance of 30Ω across 240V, 50Hz supply. The reading of voltmeter across the coil is 180V and across the resistance is 130V. Calculate power consumed by the coil and power factor of the circuit. Draw phasor diagram for the voltages. [10]
- Q3. (a) Define Dynamic Impedance, Q Factor, Bandwidth and Power Factor. [4]
(b) A voltage of $200\angle 53.8^\circ$ is applied across two impedances $(12 + j16)$ and $(10 - j20)$ in parallel. Determine the kVA, kVAR, and kW in each branch and the power factor the whole circuit. [6]
(c) Derive the condition for resonance in parallel circuits. Represent graphically the effect of variation of frequency on susceptances. [10]
- Q4. (a) How can we find Thevenin equivalent circuit if the circuit contains dependent sources? [4]
(b) Derive the condition for Maximum transfer of power from source to load. [6]
(c) [10]
- Q5. (a) What do you understand by a balanced three phase circuit? Is there any relationship between impedances connected in star and delta in a balanced system? [4]
(b) Three coils each having a resistance of 20Ω and an inductance of $0.05H$ are connected in delta to a 3phase 50Hz supply with 400V between lines. Calculate the total power absorbed and the line current. [6]

- (c) Derive the relation between voltages and currents in a star connection and draw the complete phasor diagram. [10]
- Q6. (a) Define - Reluctance, Permeance, leakage flux and useful flux. [4]
 (b) Compare electrical and Magnetic circuits. [6]
 (c) A ring has a diameter of 21cm and a cross sectional area of 10cm^2 . The ring is made up of semicircular sections of cast iron and cast steel with each joint having a reluctance equal to an air gap of 0.2mm. Find the ampere turns required to produce a flux of $8 \times 10^{-4}\text{Wb}$. The relative permeabilities of cast steel and cast iron are 800 and 166 respectively. [10]
- Q7. (a) Write formulas for coefficient of self inductance, mutual inductance, coefficient of coupling and induced e.m.f. [4]
 (b) Two coils of inductances 4 and 6 henry are connected in parallel. If their mutual inductance is 3 henry, calculate the equivalent inductance of the combination if (i) mutual inductance assists the self inductance (ii) mutual inductance opposes the self inductance. [6]
 (c) Explain the construction and working of a Permanent Magnet Moving coil instrument. Is it suitable for both ac and dc? Justify your answer. [10]

*****20/16/2013*****

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(NON-COLLEGIATE EXAMINATION)

CLASS: BE
BRANCH: EEE

SEMESTER : NC
SESSION : MO/17

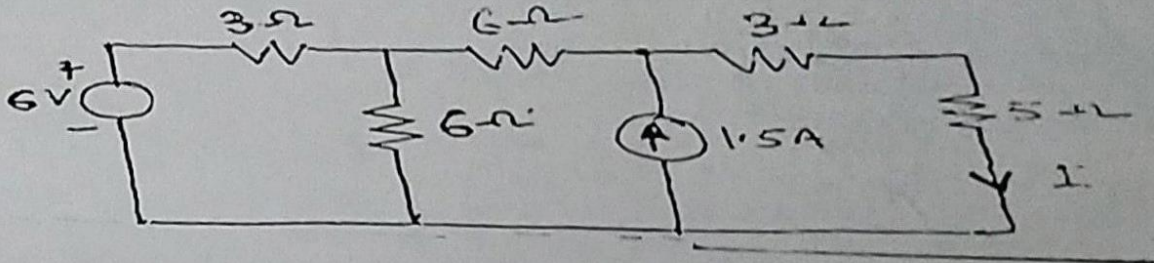
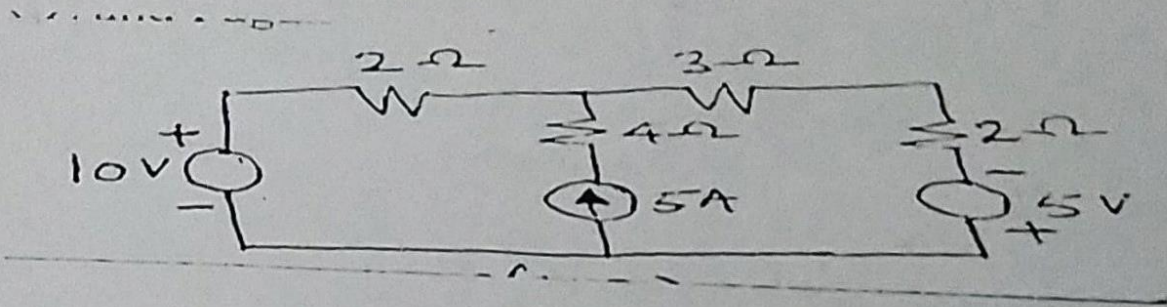
SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING
TIME: 3. hours

FULL MARKS: 100

INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
 2. Candidates may attempt all questions.
 3. The missing data, if any, may be assumed suitably.
 4. Before attempting the question paper, be sure that you have got the correct question paper.
 5. Successful candidates are entitled to 'C' grade only.
-

- Q.1(a) What are ideal voltage and current sources? Explain. [4]
Q.1(b) Obtain an expression for converting star to delta connection. [6]
Q.1(c) Use nodal analysis to find different currents. [10]
- Q.2(a) What is a power triangle? [4]
Q.2(b) Two wires carrying currents i and i join a third wire and this then carries the sum of these two currents. $i_1 = 10 \sin \omega t$ & $i_2 = 15 \sin(\omega t + 45^\circ) + \cos 3 \omega t$. What will be the equation of the current i in the third wire? What will be reading of an a.c. ammeter when connected in each of the lines with the above currents flowing. [6]
Q.2(c) An inductive coil having a resistance of 15 ohm takes a current of 4A when connected to a 100-V, 60-Hz supply. If the coil is connected to a 100-V, 50-Hz supply, calculate (i) the current, (ii) the power (iii) the p.f. [10]
- Q.3(a) What are the different methods of solving a.c parallel circuits? [4]
Q.3(b) Plot the variation of current, X_L , X_C , R , Z with frequency in a series RLC circuit. [6]
Q.3(c) A coil having a resistance of 5 ohms and an inductance of 0.1H is connected in series with a condenser of capacitance 50 μF . A constant alternating voltage of 200V is applied to the circuit. At what value of the frequency will the current be maximum? Calculate this current and the voltage across coil and across capacitor for this frequency. Find also the voltage magnification at resonance. [10]
- Q.4(a) State and explain Thevenin's theorem. [4]
Q.4(b) State and explain Maximum power transfer Theorem for a.c. Circuits. [6]
Q.4(c) Find the current I in the figures shown below using Superposition theorem. [10]
- Q.5(a) What are the advantages of 3 phase system over single phase system? [4]
Q.5(b) Derive the relations between line and phase quantities of balanced 3 phase system in starconnection. [6]
Q.5(c) A balanced delta connected load is supplied from a symmetrical three phase 400V system. The current in each phase is 30A and leads 30° behind the phase voltage find the active, reactive and apparent powers. [10]
- Q.6(a) Give examples showing diagrams of (a) series, (b) parallel and (c) series and parallel magnetic circuits. [4]
Q.6(b) What is the analogy between electric circuits and magnetic circuits? [6]
Q.6(c) A magnetic core, in the form of a closed ring, has a mean length of 20 cm and a cross-section of 1 sq.cm . The relative permeability of iron is 2400. What direct current will be needed in a coil of 2000 turns uniformly wound round the ring to create a flux of 0.2mWb in the iron? [10]
- Q.7(a) What are self and mutual inductances? [4]
Q.7(b) What are the methods of providing different types of torques? [6]
Q.7(c) Explain with the help of neat diagram the principle and working of a MI type of instrument. [10]



.....22-05-2018 NC.....

CLASS: BE
BRANCH: ALL

SEMESTER : BL
SESSION : MO/16

SUBJECT: EE2201 PRINCIPLES OF ELECTRICAL ENGINEERING

TIME: 03:00

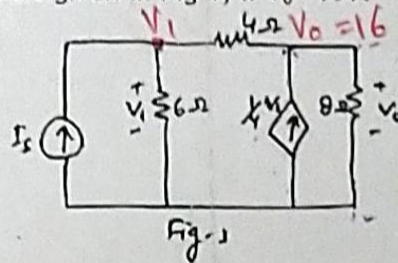
FULL MARKS: 100

INSTRUCTIONS:

1. The question paper contains 7 questions each of 20 marks and total 140 marks.
2. Candidates may attempt any 5 questions maximum of 100 marks.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.
6. Nothing should be written on front or back of the question paper except tick marking.

- Q.1(a) Differentiate between (i) mesh and loop (ii) active and passive elements. [4]
(b) State and explain Kirchhoff's current law and voltage law. [6]
(c) Determine the current I_s in the circuit given in Fig-1, if $V_0 = 16V$. [10]

$$\begin{aligned} -I_s + \frac{8}{6} + \frac{8-16}{4} &= 0 \\ -I_s + \frac{8}{6} - 2 &= 0 \\ I_s &= \frac{8}{6} - 2 \end{aligned}$$



$$\begin{aligned} -I_s + \frac{V_1}{6} + \frac{V_1 - 16}{4} &= 0 \\ -I_s + \frac{V_1}{6} + \frac{V_1}{4} - \frac{16}{4} &= 0 \\ \frac{16 - V_1}{4} - \frac{V_1}{4} + \frac{16}{8} &= 0 \\ 4 - \frac{V_1}{4} - \frac{V_1}{4} + 2 &= 0 \\ 6 - \frac{V_1}{2} &= 0 \\ V_1 &= 12 \end{aligned}$$

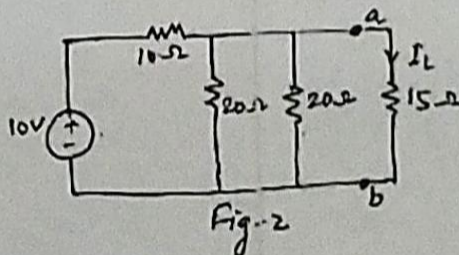
- Q.2(a) Define form factor and peak factor of an alternating quantity. [4]
(b) Draw the following waves on a suitable scale: [6]
 $v = 282.8 \sin(360 \times 25t)$ and
 $i = 28.28 \sin(360 \times 25t - 60^\circ)$

Determine (i) RMS value of voltage and average value of current (ii) periodic time and frequency (iii) instantaneous value of voltage, when instantaneous value of current is $-2.5A$.

- (c) When a resistor and an inductor in series are connected to 240V, 50Hz AC mains, a current of 3A flows lagging 37° behind the supply voltage, while the voltage across the inductor is 171V. Determine the resistance of the resistor and the resistance and reactance of the inductor. Draw the phasor diagram of the circuit and impedance triangle. [10]

- Q.3(a) Define power factor? Discuss the practical importance of power factor. [4]
(b) Define Q-factor for series resonant circuit and express it in terms of circuit parameters. [6]
(c) A parallel circuit has two branches. One branch contains a resistance of 8Ω and inductance of $19.1mH$ in series and the other contains a resistance of 6Ω and capacitor of capacitance $601.55F$ in series. This parallel circuit is connected across a supply voltage of 240V, 50Hz. Determine the (i) current drawn by each branch (ii) total current drawn from the mains and (iii) power factor of the whole circuit. Draw the phasor diagram. [10]

- Q.4(a) State and explain Thevenin's theorem. [4]
(b) Determine the current I_L through the 15Ω resistor in the network given in Fig-2, using Norton's theorem. [6]



$$\begin{aligned} R_{th} &= R_N = 5\Omega \\ I_{sc} &= 1A \\ I_L &= \frac{1 \times 5}{5+15} \end{aligned}$$

- (c) State and prove the maximum power transfer theorem for AC circuits. [10]

- Q.5(a) Describe the basic features of balanced three-phase system. [4]
 (b) Each phase of a delta-connected load has a resistance of 25Ω , an inductance of $0.15H$ and a capacitance of $120\mu F$. The load is connected across a $400V$, $50Hz$ three-phase supply. Determine the line current, active power and volt-ampere reactive power. Draw the phasor diagram. [6]
 (c) Prove that the power in a three-phase balanced circuit can be deduced from the readings of two-wattmeters with the help of suitable phasor diagrams. [10]
 Discuss the nature of power factor when (i) the two wattmeters read equal and positive values (ii) the two wattmeters read equal and opposite values and (iii) one wattmeter reads zero value.
- Q.6(a) Define (i) Reluctance (ii) MMF [4]
 (b) What is hysteresis? Define coercive force and residual flux density. Sketch the hysteresis loop and indicate on it the coercive force and residual flux density. [6]
 (c) The magnetic frame shown in Fig-3 is built-up of iron of square cross section of $3cm$ side. Each air gap is $2mm$ wide. Each of the coil is wound with 1000 turns and the exciting current is $1A$. The relative permeability of path A and path B are 1000 and 1200 , respectively. Calculate (i) reluctance of path A, path B and air gaps (ii) total reluctance (iii) MMF produced (iv) flux set up in the circuit. [10]

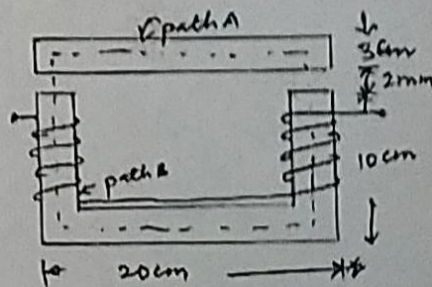


Fig-3

- Q.7(a) Explain the dot rule. [4]
 (b) Explain the concept of voltmeter and ammeter. [6]
 (c) Find the drop across the capacitor and the resistor for the circuit given in Fig-4. [10]

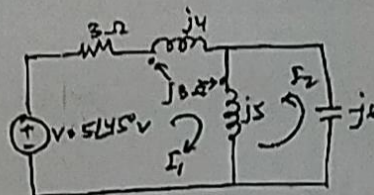


Fig-4

*****06/12/16*****M